

交通大数据

描述性统计分析与假设检验



描述性统计分析与假设检验

□ 学习目标

- 描述性统计分析
- 基本统计分布
- 置信区间
- 单样本、多样本假设检验



描述性统计分析与假设检验

□ 学习目标

- 描述性统计分析
- 基本统计分布
- 置信区间
- 单样本、多样本假设检验



描述性统计分析

Table 7. Estimation results of EVT models.

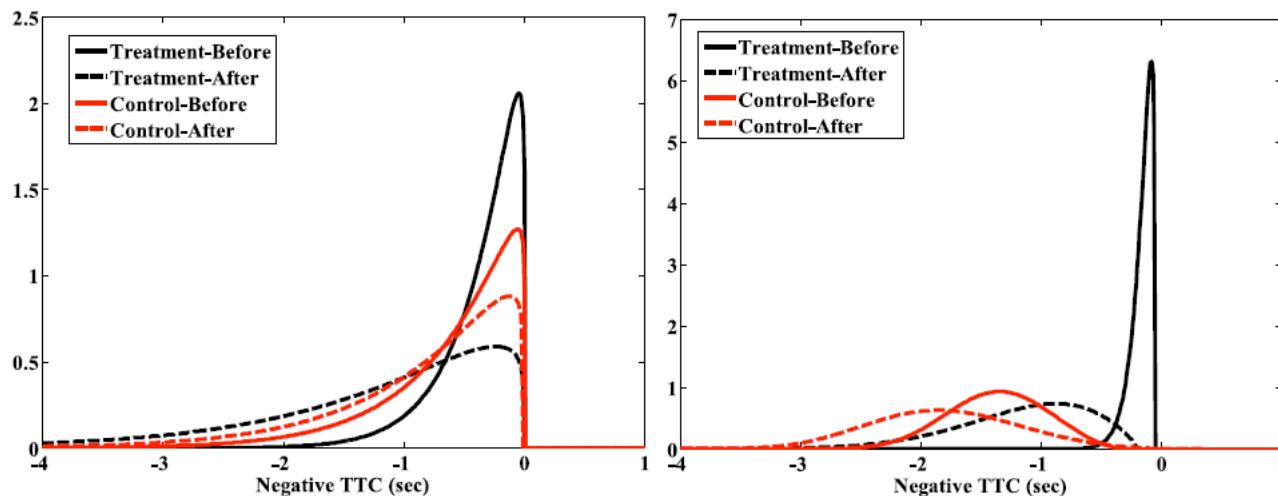
定量

- 数值
- 通过
- 样本

Intersections	Condition	Location parameter (μ)				Scale parameter (σ)		Shape parameter (ξ)	
		Mean (β_0)	Ste.	Mean (β_1)	Ste.	Mean	Ste.	Mean	Ste.
T1	Treatment-Before	-0.389	0.067	-	-	0.354	0.018	-0.906	0.105
	Treatment-After	-1.307	0.290	-	-	1.127	0.315	-0.863	0.290
	Control-Before	-0.656	0.118	-	-	0.611	0.073	-0.931	0.051
	Control-After	-0.924	0.16	-	-	0.822	0.179	-0.904	0.221
T2	Treatment-Before	-0.323	0.01	0.014	0.001	0.097	0.002	-0.820	0.046
	Treatment-After	-1.439	0.16	0.181	0.074	0.61	0.076	-0.564	0.120
	Control-Before	-1.821	0.134	0.132	0.030	0.485	0.051	-0.337	0.074
	Control-After	-2.464	0.193	0.406	0.070	0.605	0.08	-0.256	0.150
T3	Treatment-Before	-0.303	0.014	0.01	0.001	0.179	0.004	-0.802	0.054
	Treatment-After	-2.51	0.166	0.631	0.151	0.687	0.085	0.364	0.136
	Control-Before	-1.727	0.086	0.135	0.018	0.42	0.039	-0.299	0.071
	Control-After	-2.1	0.09	0.416	0.058	0.536	0.061	-0.258	0.124

定性

- 图元
- 列表



(a) GEV distributions for intersection T1

(b) GEV distributions for intersection T2

定量描述

□ 相对位置 (relative standing)

- 百分位 (percentile)
- 90分位数：90%的观测数据小于该值
- 四分位 (quartile)
- 把观测数据**一分为四**的百分位数
- 第一 (first or lower) 四分位数：25%的观测数据小于该值
- 第二 (second or middle) 四分位数：50%的观测数据小于该值
- 第三 (third or upper) 四分位数：75%的观测数据小于该值
- 四分位距 (interquartile range)：第一、三四分位数之差



定量描述

□ 集中趋势 (central tendency)

- 中位数 (median) : 50分位数
- 平均值 (arithmetic mean)

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

Population Mean

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

Sample Mean

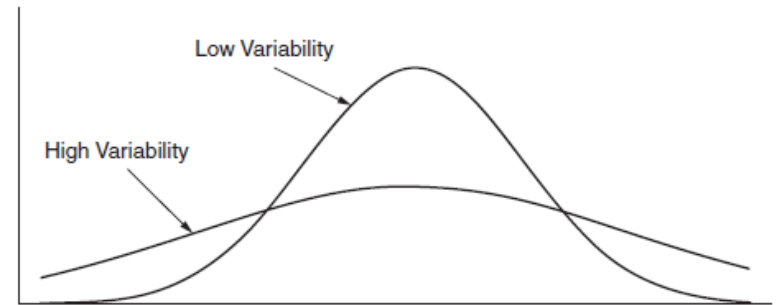
- 众数 (mode) : 出现最多次的观测值



定量描述

□ 变化性 (variability)

- 定义：衡量数据**离散性**的统计指标
- 四分位距 (interquartile range)
- 区间 (range)：最大、最小观测值之差
- 方差 (variance) 或标准偏差 (standard deviation)



$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

**Population Standard
Deviation**

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{X})^2}{n-1}}$$

**Sample Standard
Deviation**

定量描述

□ 变化性 (variability)

- 只看标准偏差够吗?
- 标准偏差是数据离散性的**绝对**衡量指标, 未考虑数据**量级**
- 如何解决?
- 变异系数 (coefficient of variation)

$$CV = \frac{s}{\bar{X}}$$



定量描述

□ 定量描述性统计分析案例1

TABLE 1.1

Descriptive Statistics for Speeds on Indiana Roads

Statistic	Value
<i>N</i> (number of observations)	1296
Mean	58.86
Std. deviation	4.41
Variance	19.51
CV	0.075
Maximum	72.5
Minimum	32.6
Upper quartile	61.5
Median	58.5
Lower quartile	56.4



定量描述

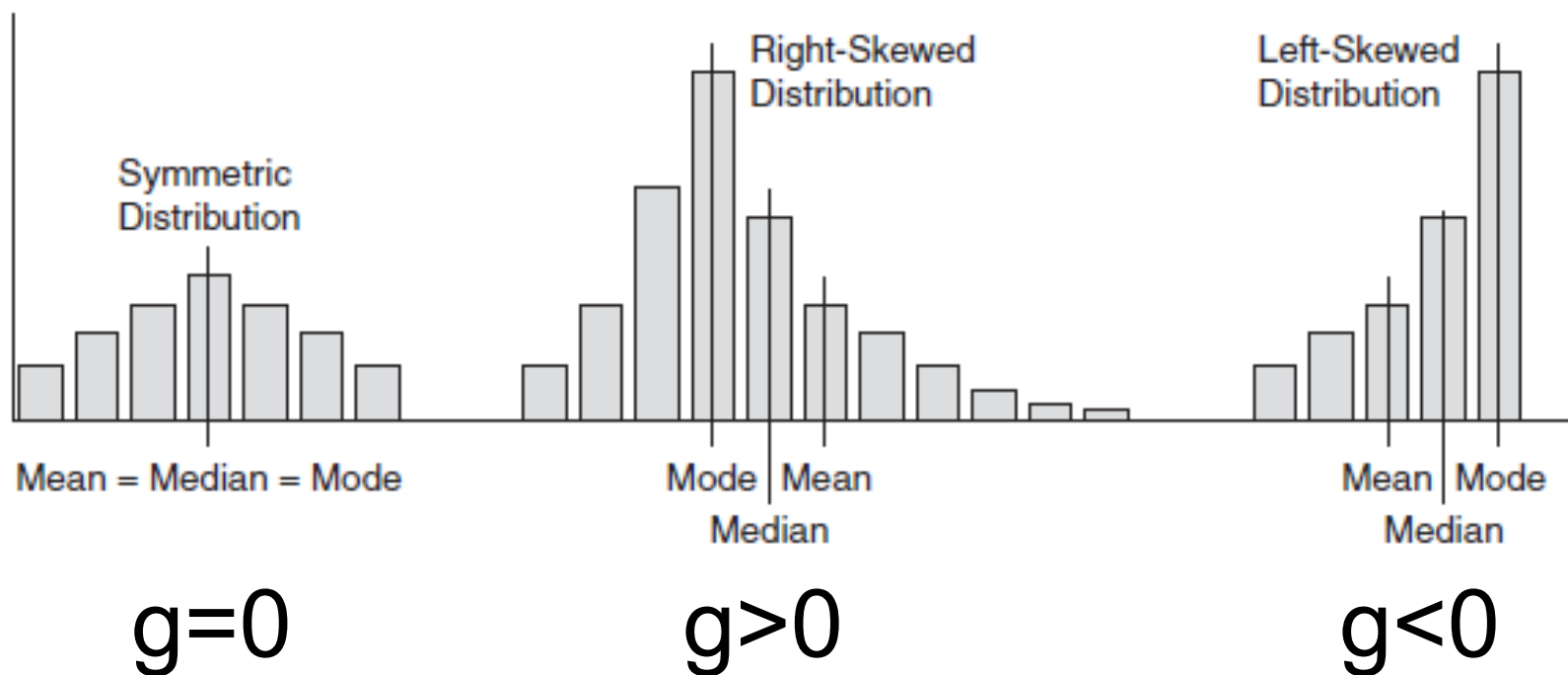
□ 偏度 (skewness)

- 描述数据分布**偏斜方向和程度**
- 描述数据分布**不对称程度**
- 描述概率分布密度曲线相对于平均值不对称程度
- 又叫**第三中心距** (third central moment)
- 当概率分布向右延伸，即右侧有“长尾巴”，此概率分布是**右偏斜**的 (right-skewed) 或正偏斜的 (positively skewed) ；
- 当概率分布向左延伸，即左侧有“长尾巴”，此概率分布是**左偏斜**的 (left-skewed) 或负偏斜的 (negatively skewed) 。



定量描述

□ 偏度 (skewness)

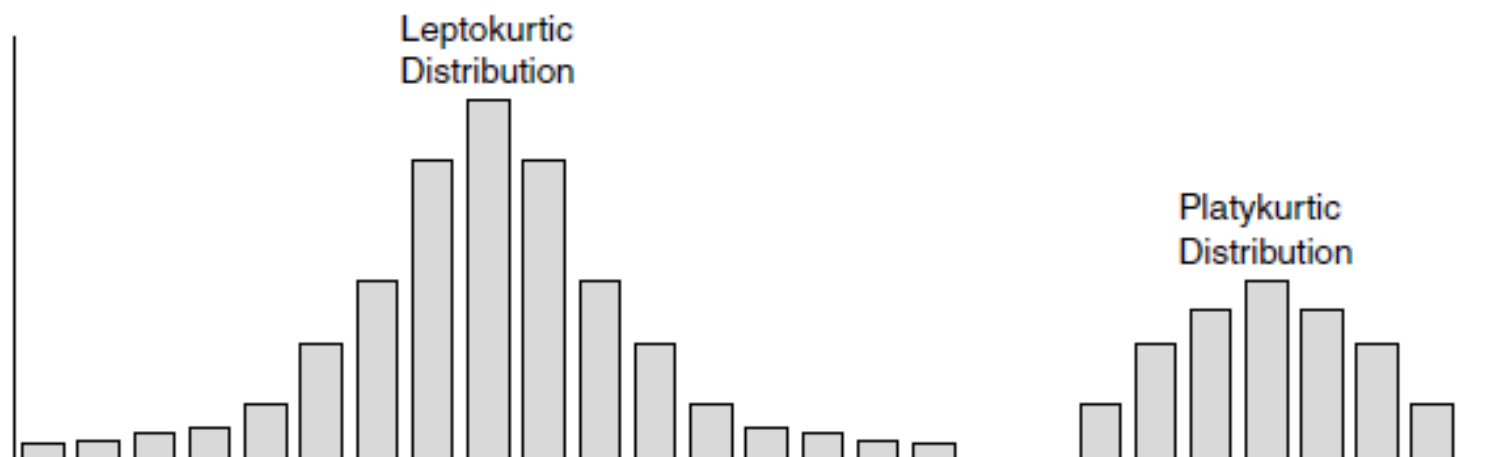


$$g_1 = \frac{m_3}{(m_2 \sqrt{m_2})} \quad m_3 = \sum_{i=1}^n (x_i - \bar{X})^3 / n \quad m_2 = \sum_{i=1}^n (x_i - \bar{X})^2 / n$$

定量描述

□ 峰度 (kurtosis)

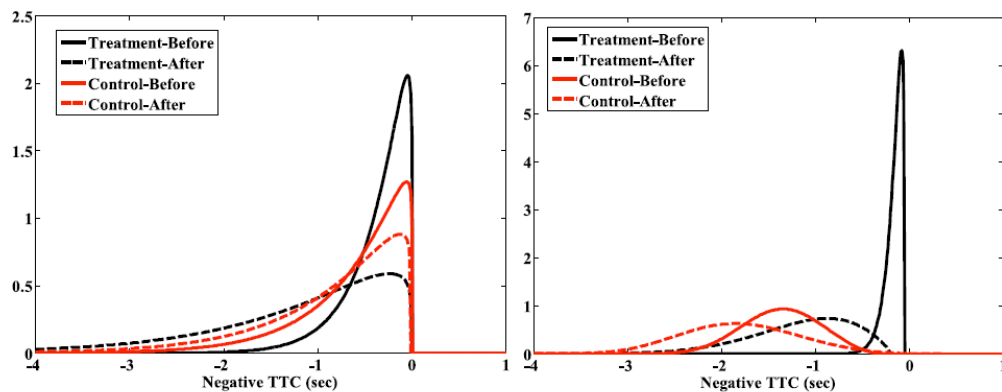
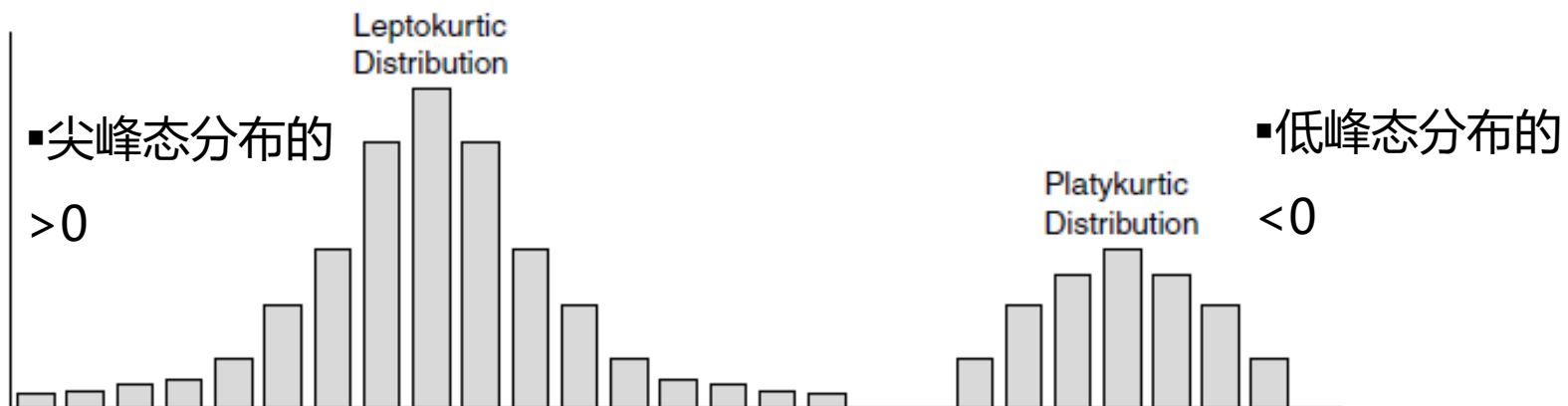
- 描述数据分布的平坦 (flatness) 或拔尖 (peakness) 程度
- 又叫**第四中心距** (fourth central moment)



$$g_2 = \gamma_2 - 3 = (m_4 / m_2^2) - 3 \quad m_4 = \sum_{i=1}^n (x_i - \bar{X})^4 / n$$

定量描述

峰度 (kurtosis)



(a) GEV distributions for intersection T1

(b) GEV distributions for intersection T2

定量描述

□ 定量描述性统计分析案例2

- 经统计分析，南京市区和郊区道路车速分布的偏度分别为-0.67 和 0.056。问，南京市区和郊区道路车速分布分别是右偏斜的还是左偏斜的？
- 经统计分析，南京市区和郊区道路车速分布的峰度分别为0.49 和-2.74。问，南京市区和郊区道路车速分布分别是尖峰态的还是低峰态的？



定量描述

□ 相关性 (association)

- 之前的描述指标仅针对**单一变量**，未描述**两个变量间的相关关系**
- 相关性 (correlation) 指标衡量了两个变量间的**线性关联**
- 协方差 (covariance)

$$\text{COV}_p(X, Y) = \frac{\sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)}{N} \quad \text{COV}_s(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})}{n-1}$$

- 大于 (小于) 0, 两个变量延同 (反) 方向变化
- 等于0, 两个变量线性无关



定量描述

□ 相关性 (association)

- 皮尔逊相关系数 (Pearson product-moment correlation parameter)

$$\rho = \frac{\text{COV}_p(X, Y)}{\sigma_X \sigma_Y}$$

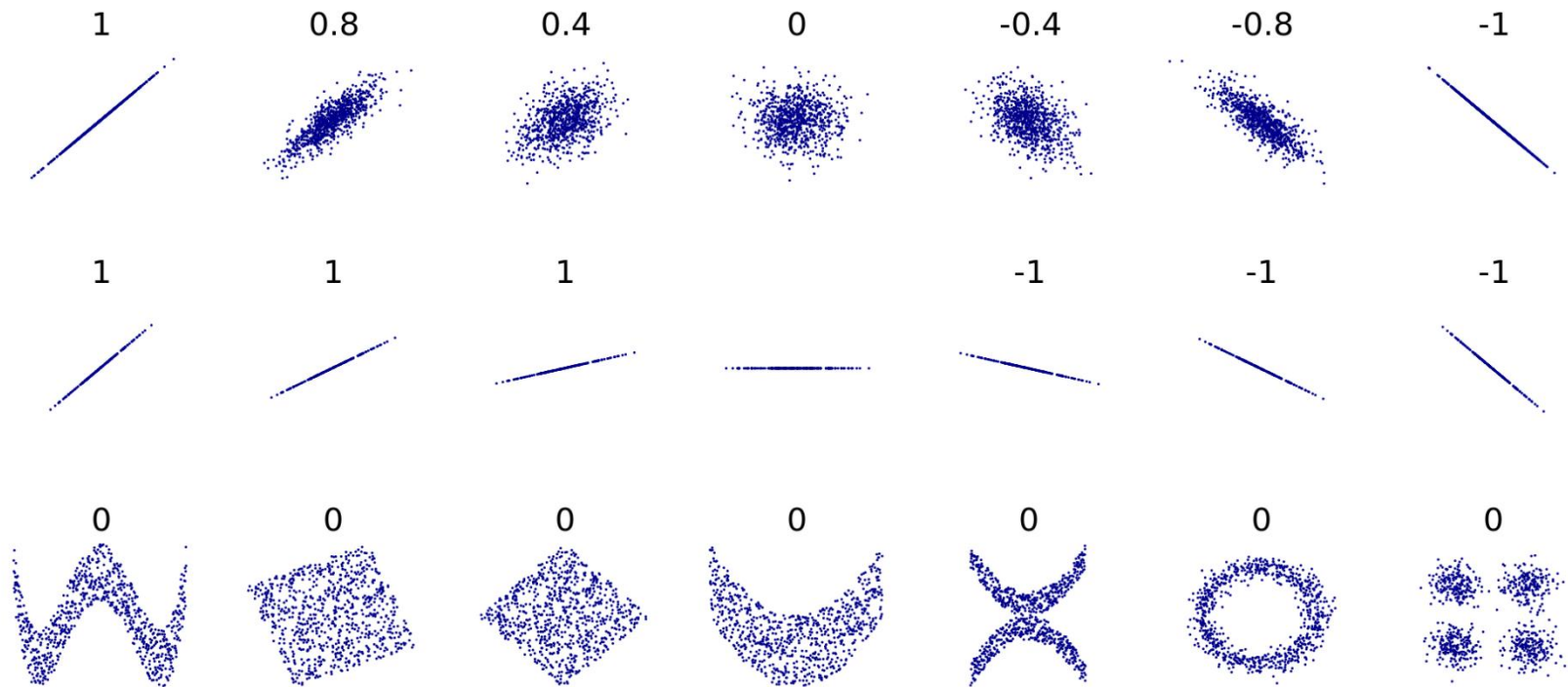
$$r = \frac{\text{COV}_s(X, Y)}{s_X s_Y}$$

- 始终取值在 $[-1, 1]$



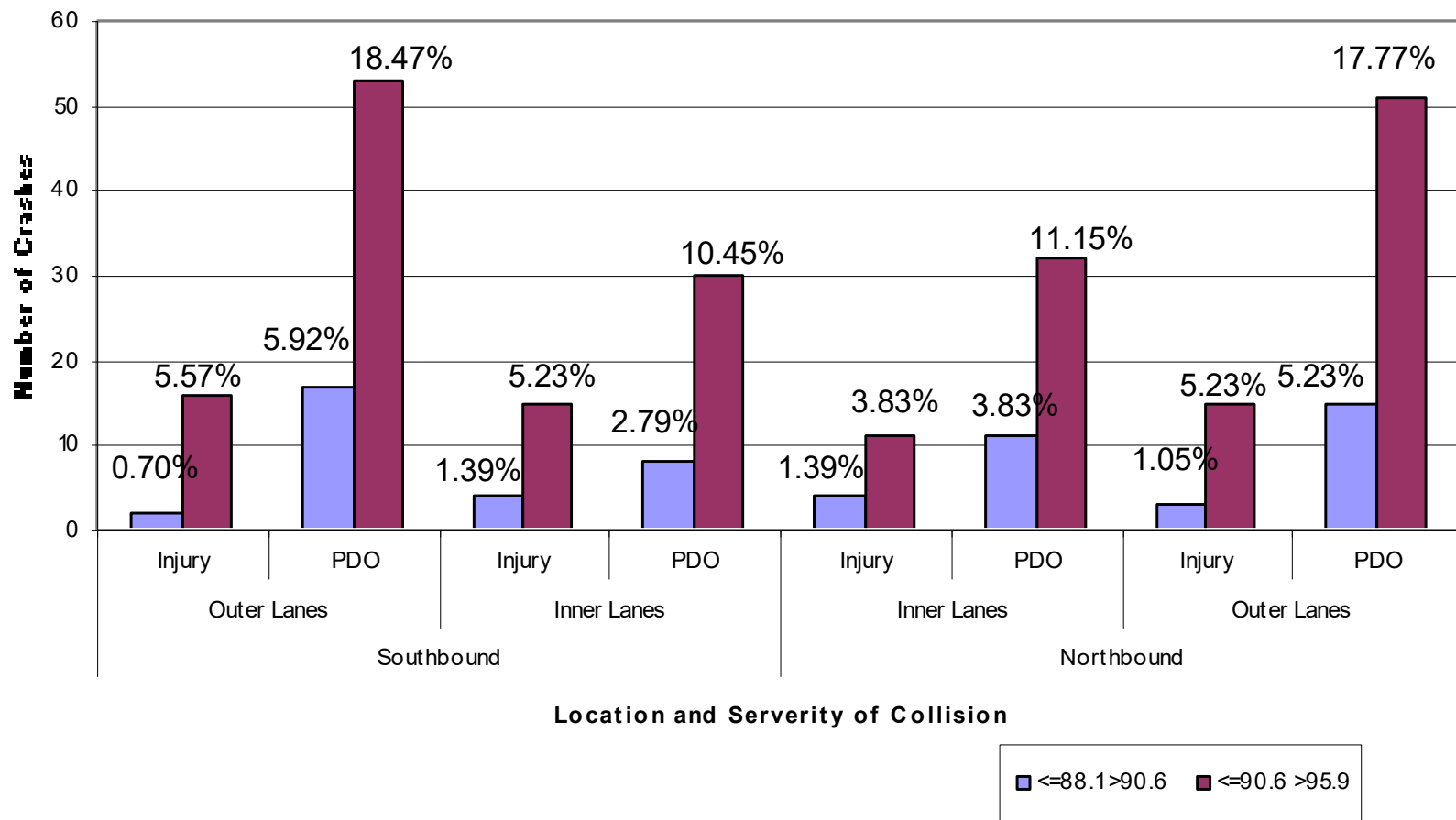
定量描述

□ 相关性 (association)



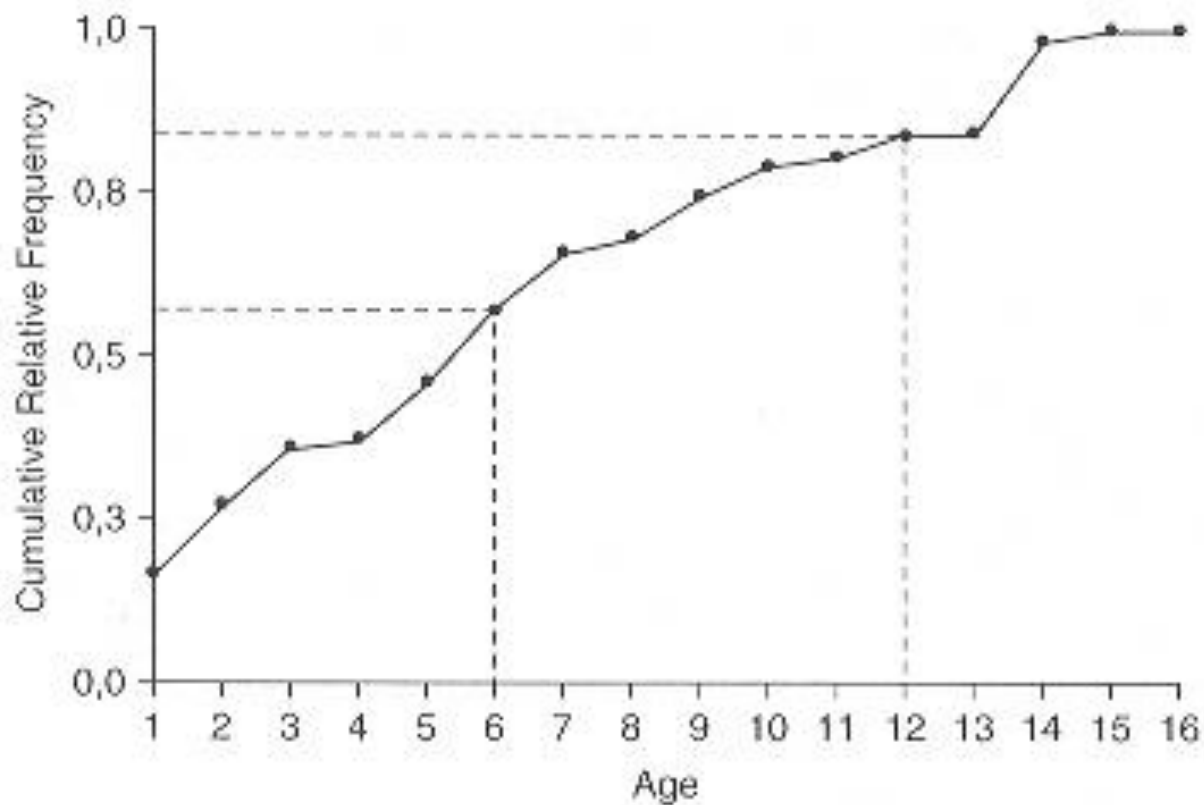
定性描述

直方图 (histogram)



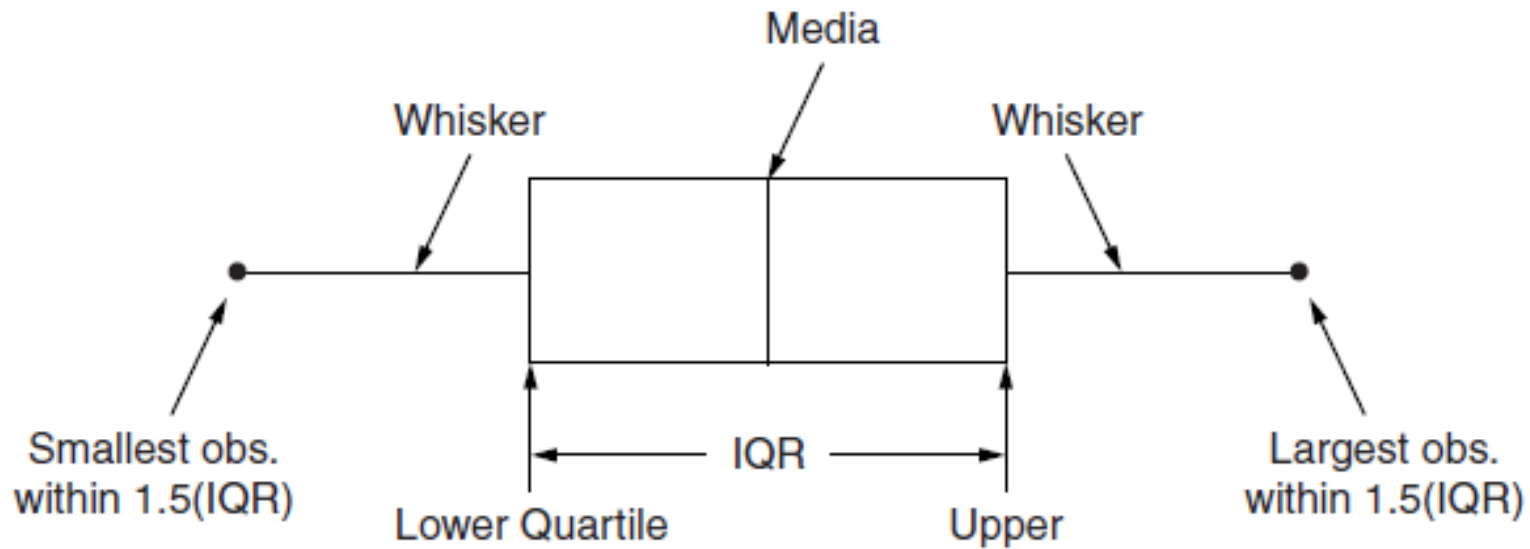
定性描述

□ 累积曲线图 (ogive)



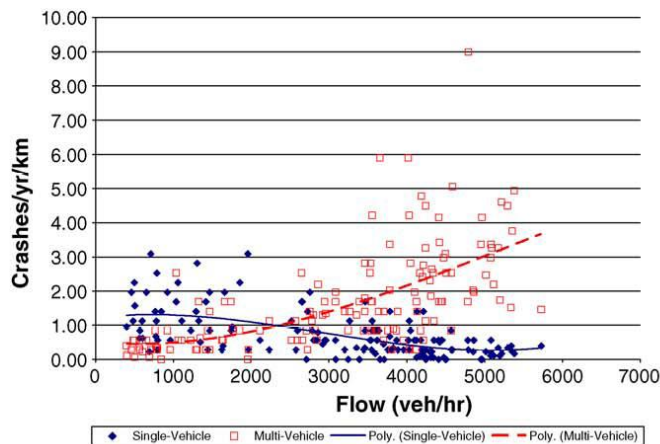
定性描述

□ 箱型图 (box plot)



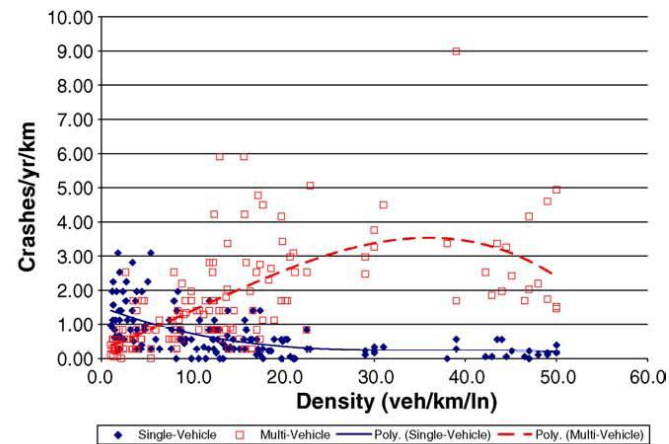
定性描述

散点图 (scatter plot)



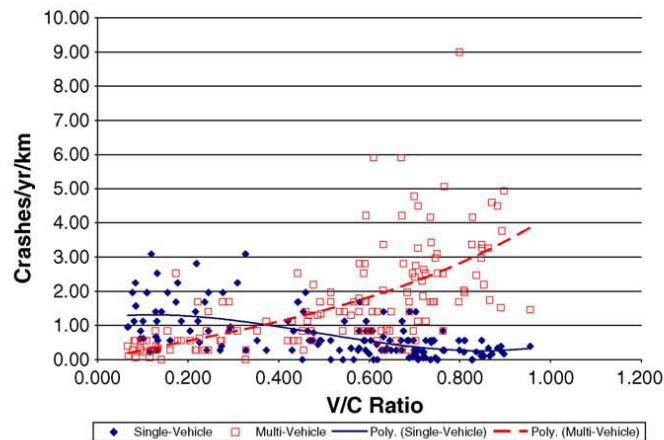
(A)

Crash-Flow Relationship



(B)

Crash-Density Relationship

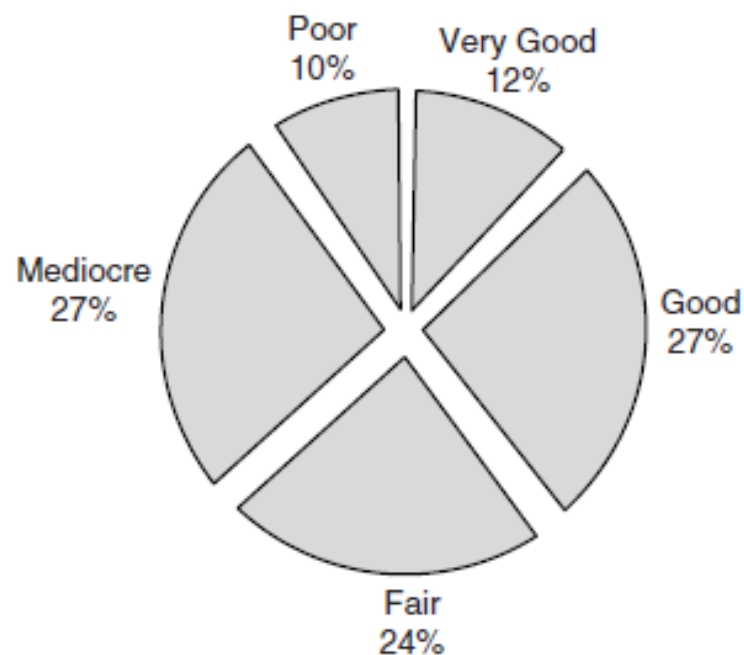
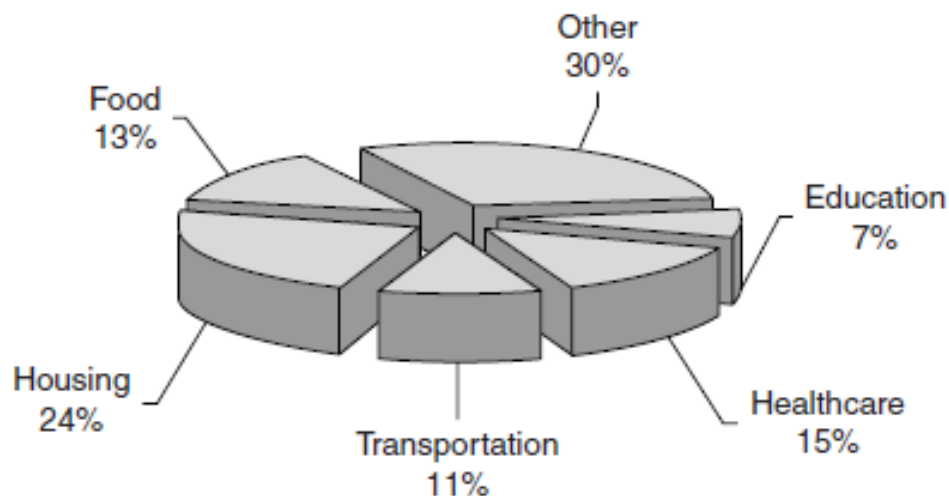


(C)

Crash-V/C Ratio Relationship

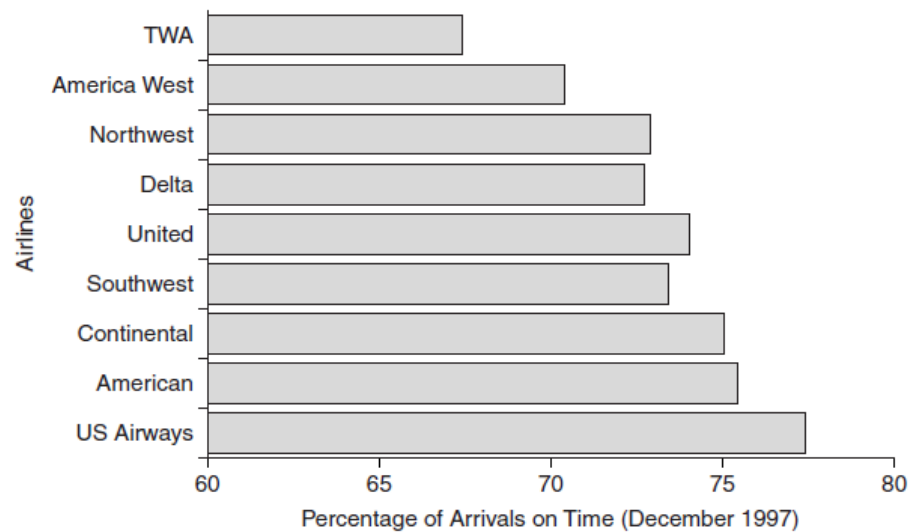
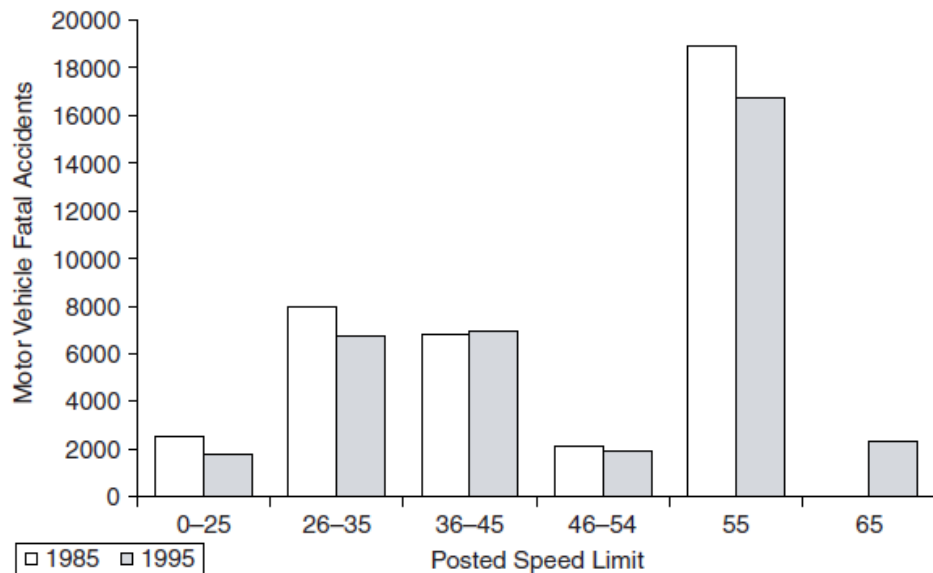
定性描述

□ 饼图 (pie chart)



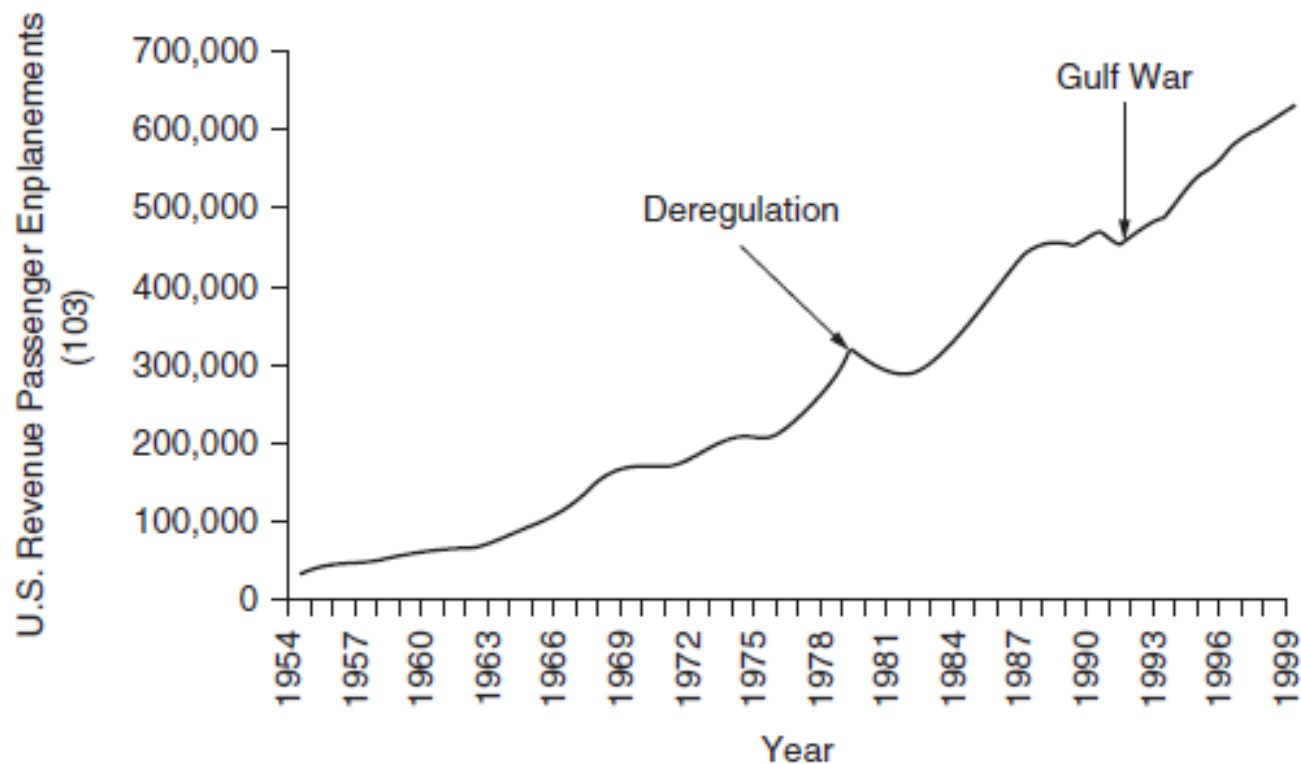
定性描述

□ 条形图 (bar chart)



定性描述

□ 线性图 (line chart)



定性描述

□ 表格 (table)

Crash Severity / Flow Range	< 5,000	5,000-9,999	$\geq 10,000$
Fatal	10	12	15
Non-Fatal Injury	100	120	135
PDO	550	700	900

描述性统计分析与假设检验

□ 学习目标

- 描述性统计分析
- 基本统计分布
- 置信区间
- 单样本、多样本假设检验



基本统计分布

□ 两类概率分布

- 连续 vs. 离散
- 二项分布, 泊松分布, 正态分布

□ 两类描述方式

- 概率密度函数 (probability density function, PDF)
- 概率质量函数 (probability mass function, PMF)
- 累积分布函数 (cumulative distribution function, CDF)



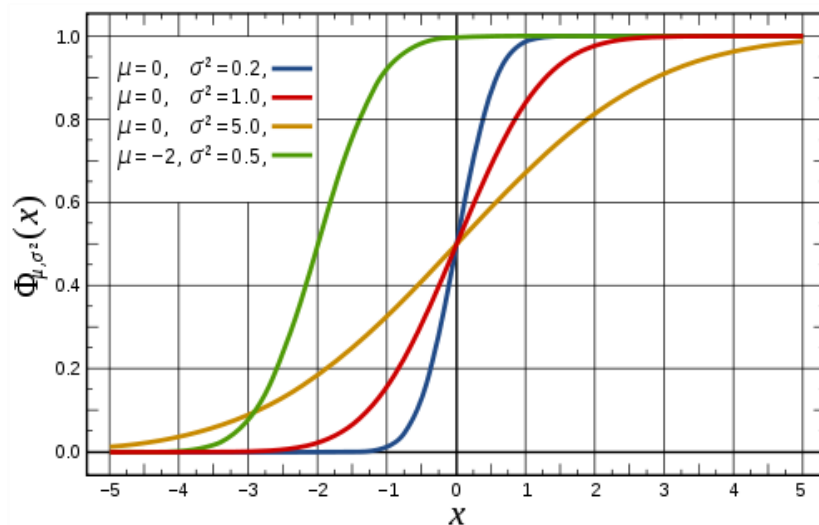
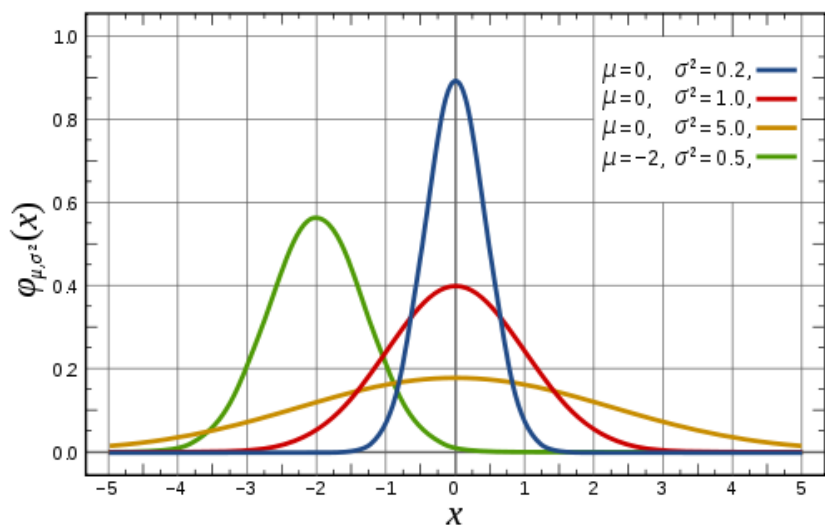
正态分布

□ PDF有两个参数

- 期望 μ 和方差 σ^2

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- 如果 $X \sim N(\mu, \sigma^2)$, 那么 $(X - \mu) / \sigma \sim N(0, 1)$



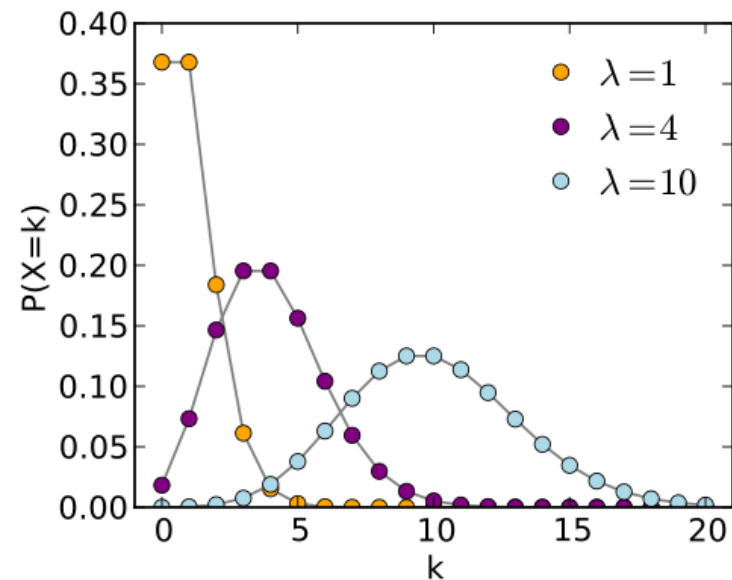
泊松分布

□ PMF有一个参数

- 单位时间内事件平均发生率 λ

$$f(k; \lambda) = \frac{e^{-\lambda} \lambda^k}{k!}$$

- 事件在单位时间内发生 k 次的概率
- 事件**独立**发生
- 期望 = 方差 = λ



描述性统计分析与假设检验

□ 学习目标

- 描述性统计分析
- 基本统计分布
- 置信区间
- 单样本、多样本假设检验



置信区间

□ 点估计 (point estimate)

- 通过一个**样本参数值**对总体参数进行估计

□ 区间估计 (interval estimate)

- 通过一个**置信区间**对总体参数进行估计 (考虑某**置信水平**)
- 置信水平: **总体参数落在该区间的概率**
- 置信下限 (lower confidence limit)
- 置信上限 (upper confidence limit)



置信区间

□ 总体 σ 已知, μ 未知

- 中心极限定理 (central limit theorem) : 若总体均值和标准偏差分别表示为 μ 和 σ , 当独立取样足够多时, 样本均值 $\bar{X}$ 近似服从均值为 μ 、标准偏差为 $\frac{\sigma}{\sqrt{n}}$ 的正态分布。

$$Z^* = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$



置信区间

□ 总体 σ 已知, μ 未知

- 标准正态分布变量 Z^* 落在区间 $[-1.96, 1.96]$ 的概率是0.95。

$$P\left(-1.96 < \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < 1.96\right) = 0.95$$

$$P\left(-1.96 \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < 1.96 \frac{\sigma}{\sqrt{n}}\right) = 0.95$$

$$P\left(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}\right) = 0.95$$



置信区间

□ 总体 σ 已知, μ 未知

- 总体均值 μ 落入区间

$$(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}})$$

的概率为95%; 该区间称为 μ 的95%置信区间。

- μ 的 $(1 - \alpha) \times 100\%$ 置信区间为

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- $z_{\alpha/2}$: 当标准正态分布两端曲线下面积之和为 α 时 z 的取值



Normal Distribution

continued

z	$f(z)$	$F(z)$	$1 - F(z)$	z	$f(z)$	$F(z)$	$1 - F(z)$
-----	--------	--------	------------	-----	--------	--------	------------

continued

置信区间

□ 总体 σ 已知, μ 未知——案例

A 95% confidence interval is desired for the mean vehicular speed on Indiana roads (see Example 1.1 for more details). First, the assumption of normality is checked; if this assumption is satisfied we can proceed with the analysis. The sample size is $n = 1296$, and the sample mean is $\bar{X} = 58.86$. Suppose a long history of prior studies has shown the population standard deviation as $\sigma = 5.5$. Using Equation 2.4, the confidence interval can be obtained:

$$\bar{X} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 58.86 \pm 1.96 \frac{5.5}{\sqrt{1296}} = 58.86 \pm 0.30 = [58.56, 59.16]$$

The result indicates that the 95% confidence interval for the unknown population parameter μ consists of lower and upper bounds of 58.56 and 59.16. This suggests that the true and unknown population parameter would lie somewhere in this interval about 95 times out of 100, on average.



置信区间

□ 总体 σ 和 μ 均未知

- 在实际数据分析中，总体标准偏差获取较难；因此， μ 的 $(1 - \alpha) \times 100\%$ 置信区间为

$$\bar{X} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

- $t_{\alpha/2}$ ：当 $n - 1$ 自由度的 t 分布两端曲线下面积之和为 α 时 t 的取值



ν	α						
	0.1	0.05	0.025	0.01	0.005	0.0025	0.001
1	3.078	6.314	12.706	31.821	63.657	318.309	636.619
2	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
35	1.306	1.690	2.030	2.438	2.724	3.340	3.591
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
45	1.301	1.679	2.014	2.412	2.690	3.281	3.520
50	1.299	1.676	2.009	2.403	2.678	3.261	3.496
100	0.290	1.660	1.984	2.364	2.626	3.174	3.390
∞	1.282	1.645	1.960	2.326	2.576	3.091	3.291



置信区间

□ 总体 σ 和 μ 均未知——案例

Continuing with the previous example, a 95% confidence interval for the mean speed on Indiana roads is computed, assuming that the population variance is not known, and instead an estimate is obtained from the data with the same value as before. The sample size is $n = 1296$, and the sample mean is $\bar{X} = 58.86$. Using Equation 2.3, the confidence interval can be obtained as $S=4.41$

$$\bar{X} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} = 58.86 \pm 1.96 \frac{4.41}{\sqrt{1296}} = [58.61, 59.10]$$



置信区间

□ 总体比例 (population proportion)

- 某特征在总体中的发生比例
- 二项分布 $B(n, p)$
- n 足够大 ($np \geq 5, nq \geq 5$) , 二项分布趋近于正态分布 $N(np, npq)$
或 $N\left(p, \frac{pq}{n}\right)$, $q = 1 - p$
- p 的 $(1 - \alpha) \times 100\%$ 置信区间为

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$



置信区间

□ 总体比例——案例

A transit planning agency wants to estimate, at a 95% confidence level, the share of transit users in the daily commute “market” (that is, the percentage of commuters using transit). A random sample of 100 commuters is obtained and it is found that 28 people in the sample are transit users. By using Equation 2.6, a 95% confidence interval for p is calculated as

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} = 0.28 \pm 1.96 \sqrt{\frac{(0.28)(0.72)}{100}} = 0.28 \pm 0.088 = [0.192, 0.368].$$

Thus, the agency is 95% confident that transit users in the daily commute range from 19.2 to 36.8%.



置信区间

□ 总体方差 (population variance)

- σ^2 的 $(1 - \alpha) \times 100\%$ 置信区间为

$$\left[\frac{(n-1)s^2}{\chi_{\alpha/2}^2}, \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2} \right]$$

- $\chi_{\alpha/2}^2$ 和 $\chi_{1-\alpha/2}^2$: 当 $n - 1$ 自由度的卡方分布两端曲线下面积分别为 $\alpha/2$ 和 $1 - \alpha/2$ 时 χ^2 的取值



置信区间

□ 总体方差 (population variance)

v	α							
	.9999	.9995	.999	.995	.99	.975	.95	.90
41	15.48	17.54	18.58	21.42	22.91	25.21	27.33	29.91
42	16.07	18.19	19.24	22.14	23.65	26.00	28.14	30.77
43	16.68	18.83	19.91	22.86	24.40	26.79	28.96	31.63
44	17.28	19.48	20.58	23.58	25.15	27.57	29.79	32.49
45	17.89	20.14	21.25	24.31	25.90	28.37	30.61	33.35
46	18.51	20.79	21.93	25.04	26.66	29.16	31.44	34.22
47	19.13	21.46	22.61	25.77	27.42	29.96	32.27	35.08
48	19.75	22.12	23.29	26.51	28.18	30.75	33.10	35.95
49	20.38	22.79	23.98	27.25	28.94	31.55	33.93	36.82
50	21.01	23.46	24.67	27.99	29.71	32.36	34.76	37.69
60	27.50	30.34	31.74	35.53	37.48	40.48	43.19	46.46
70	34.26	37.47	39.04	43.28	45.44	48.76	51.74	55.33
80	41.24	44.79	46.52	51.17	53.54	57.15	60.39	64.28
90	48.41	52.28	54.16	59.20	61.75	65.65	69.13	73.29
100	55.72	59.90	61.92	67.33	70.06	74.22	77.93	82.36

v	α							
	.10	.05	.025	.01	.005	.001	.0005	.0001
60	74.40	79.08	83.30	88.38	91.95	99.61	102.69	109.50
70	85.53	90.53	95.02	100.43	104.21	112.32	115.58	122.75
80	96.58	101.88	106.63	112.33	116.32	124.84	128.26	135.78
90	107.57	113.15	118.14	124.12	128.30	137.21	140.78	148.63
100	118.50	124.34	129.56	135.81	140.17	149.45	153.17	161.32



置信区间

□ 总体方差——案例

A 95% confidence interval for the variance of speeds on Indiana roads is desired. With a sample size of 100 and a variance of 19.51 mph², and using the values from the χ^2 table (Appendix C, Table C.3), one obtains $\chi^2_{\alpha/2} = 129.56$ and $\chi^2_{1-\alpha/2} = 74.22$. Thus, the 95% confidence interval is given as

$$\left[\frac{99(19.51)}{129.56}, \frac{99(19.51)}{74.22} \right] = [15.05, 26.02]$$

The speed variance is, with 95% confidence, between 15.05 and 26.02. Again, the units of the variance in speed are in mph².



描述性统计分析与假设检验

□ 学习目标

- 描述性统计分析
- 基本统计分布
- 置信区间
- 单样本、多样本假设检验



假设检验

□ 目的

- 判断样本与样本、样本与总体的差异（均值、方差、比例等）是由**抽样误差**引起还是**本质差别**造成的
- 显著性检验（significance test）

□ 基本原理

- 先对总体的特征做出某种**假设**，然后通过**抽样**进行统计分析，推断此假设应该被**拒绝**还是**不拒绝**
- 小概率事件实际不可能性： $P(\text{data}|\text{null hypothesis})$



假设检验

□ 基本步骤

- 提出假设检验，即**无效假设**（null hypothesis, H_0 ），和**备择假设**（alternative hypothesis, H_1 ）；选定**显著性水平**（0.05）
- 选定**检验统计量**（test statistic, Z, t, F, χ^2 ）
- 计算检验统计量及相应 P 值；若在拒绝域（rejection or critical region），拒绝 H_0 ，否则不拒绝

Critical Points of Z for Selected Levels of Significance

	Level of Significance α		
	0.10	0.05	0.01
One-tailed test	± 1.28	± 1.645	± 2.326
Two-tailed test	± 1.645	± 1.96	± 2.576



假设检验

□ 两类错误

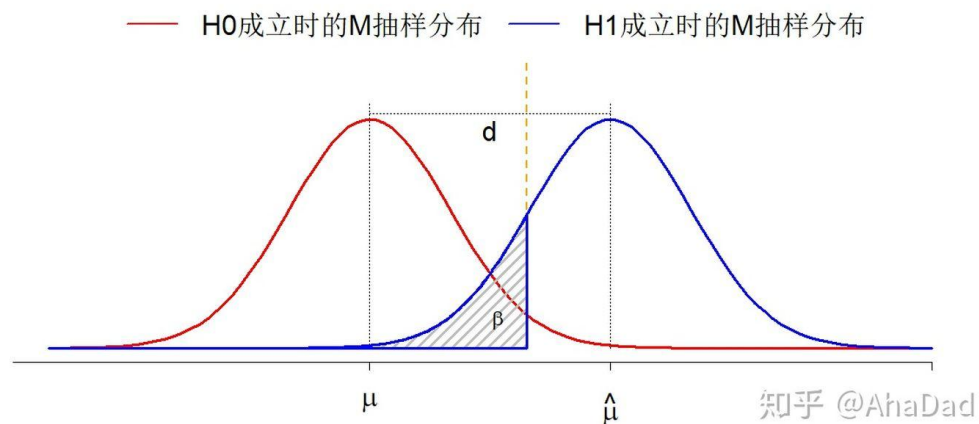
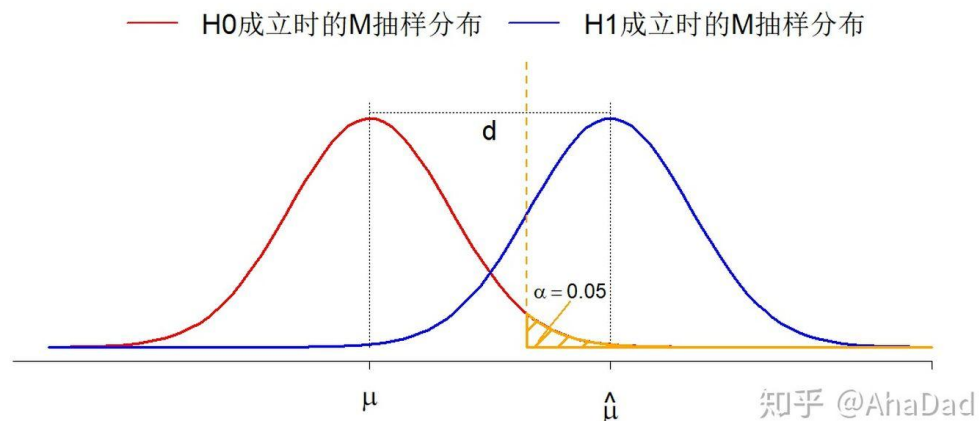
- 统计检验力
(power of the test)
- $1 - \beta$

接受或拒绝 H_0 ，都可能犯错误

I类错误——弃真错误，发生的概率为 α

II类错误——取伪错误，发生的概率为 β

检验决策	H_0 为真	H_0 非真
拒绝 H_0	犯I类错误 (α)	正确
接受 H_0	正确	犯II类错误 (β)



假设检验

□ 假设检验——案例1

Using the data from Example 2.1, a test is conducted to assess whether the mean speed on Indiana roads is 59.5 mph at the 5% significance level. The sample size is $n = 1296$, and the sample mean is $\mu = 58.86$. Suppose that numerous past studies have revealed the population standard deviation to be $\sigma = 5.5$. The parameter of interest is the population mean, and the hypotheses to be tested are

$$H_0: \mu = 59.5$$

$$H_a: \mu \neq 59.5$$

From Example 2.1, a 95% confidence interval for mean speeds is [58.56, 59.16]. Because the value 59.5 mph is not within the confidence interval, the null hypothesis is rejected. Thus, there is sufficient evidence to infer that mean speed is not equal to 59.5 mph.



假设检验

□ 假设检验——案例1

An alternative method for obtaining this result is to use the standardized test statistic presented in Equation 2.9 and follow the appropriate decision rules. The test statistic is

$$Z^* = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{58.86 - 59.5}{5.5 / \sqrt{1296}} = -3.27$$

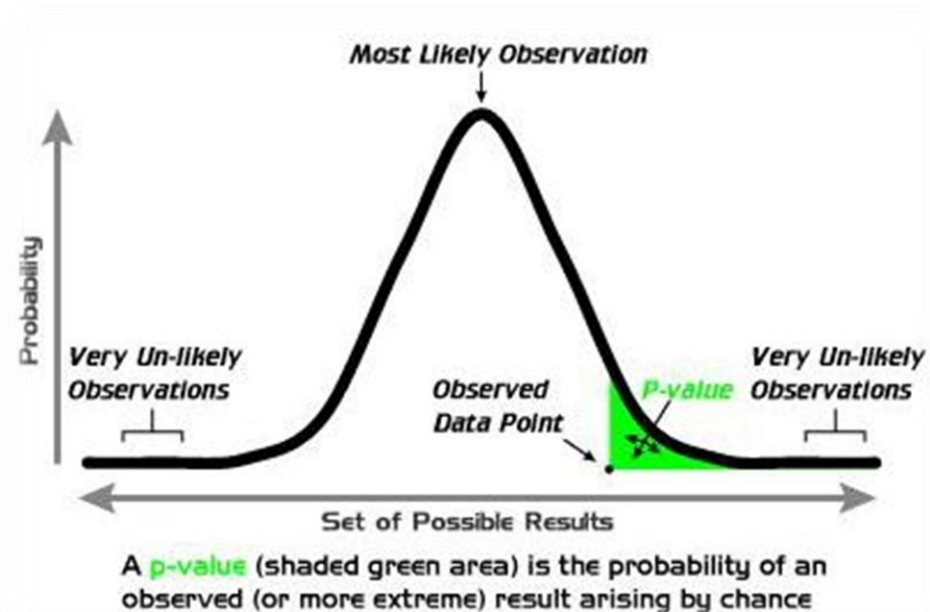
Since the test statistic $|-3.27| = 3.27$ is greater than 1.96, the critical value for a two-tailed test at the 5% level of significance, the null hypothesis is rejected. As expected, a confidence interval and the standardized test statistic lead to identical conclusions.



假设检验

□ P 值

- 拒绝 H_0 的最小显著性水平
- 接受 H_1 的统计论据
- 检验统计量越大, P 值越小, 越接受 H_1
- 若 $P \leq \alpha$, 拒绝 H_0
- 如果 $P = 0.031$, 在显著性水平 0.05 的条件下拒绝 H_0 , 在显著性水平 0.01 的条件下不拒绝 H_0



假设检验

□ 单样本检验

- 总体均值：方差未知
- 总体方差
- 总体比例

□ 多样本检验

- 两个总体均值：独立或配对样本
- 两个总体方差
- 两个总体比例



假设检验

□ 单样本总体均值：方差未知

$$Z^* = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \qquad t^* = \frac{\bar{X} - \mu}{s / \sqrt{n}}$$

□ 单样本总体方差

$$X^2 = \frac{(n-1)s^2}{\sigma^2}$$

□ 总体比例

$$Z^* = \frac{\hat{p} - p}{\sqrt{\hat{p}\hat{q}/n}}$$



假设检验

□ 两个总体均值：独立样本

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_a: \mu_1 - \mu_2 \neq 0$$

$$Z^* = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

假设检验

□ 两个总体均值：配对样本

$$H_0: \mu_d = 0$$

$$H_a: \mu_d \neq 0$$

$$t^* = \frac{\bar{X}_d - \mu_d}{\frac{s_d}{\sqrt{n_d}}}$$



假设检验

□ 两个总体方差

$$H_o: \sigma_1^2 = \sigma_2^2$$

$$H_a: \sigma_1^2 \neq \sigma_2^2$$

$$F_{(n_1-1, n_2-1)}^* = \frac{s_1^2}{s_2^2}$$



假设检验

□ 两个总体比例

$$H_0: p_1 - p_2 = 0$$

$$H_a: p_1 - p_2 \neq 0$$

$$Z^* = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

假设检验

□ 假设检验——案例2

Similar to the previous example, a test of whether the mean speed on Indiana roads is 60 mph is conducted at the 5% significance level. The sample size is $n = 1296$, and the sample mean is $\bar{X} = 58.86$. As is most often the case in practice, σ^2 is replaced with $s = 4.41$, the estimated standard deviation. The statistical hypotheses to be tested are

$$H_0: \mu = 60$$

$$H_a: \mu \neq 60$$

$$t^* = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} = \frac{58.86 - 60}{\frac{4.41}{\sqrt{1296}}} = -335$$

检验统计量非常大，拒绝无效假设；即如果平均速度是60mph，观测到这些数据的可能性极小。



假设检验

□ 假设检验——案例3

Interest is focused on whether the repeal of the NMSL had an effect on the mean speeds on Indiana roads. To test this hypothesis, 744 observations in the before period and 552 observations in the after the repeal period are used. A 5% significance level is used. Descriptive statistics show that average speeds in the before and after periods are $\bar{X}_b = 57.65$ and $\bar{X}_a = 60.48$, respectively. Further, the variances for the before and after the repeal periods are $s_b^2 = 16.4$ and $s_a^2 = 19.1$, respectively. The competing hypotheses are

$$H_0: \mu_a - \mu_b = 0$$

$$H_a: \mu_a - \mu_b \neq 0$$

$$Z^* = \frac{(\bar{X}_a - \bar{X}_b) - (\mu_a - \mu_b)}{\sqrt{\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}}} = \frac{(60.48 - 57.65) - 0}{\sqrt{\frac{19.1}{552} + \frac{16.4}{744}}} = 11.89$$



假设检验

□ 假设检验——案例3

Interest is focused on whether the repeal of the NMSL had an effect on the mean speeds on Indiana roads. To test this hypothesis, 744 observations in the before period and 552 observations in the after the repeal period are used. A 5% significance level is used. Descriptive statistics show that average speeds in the before and after periods are $\bar{X}_b = 57.65$ and $\bar{X}_a = 60.48$, respectively. Further, the variances for the before and after the repeal periods are $s_b^2 = 16.4$ and $s_a^2 = 19.1$, respectively. The competing hypotheses are

$$(\bar{X}_a - \bar{X}_b) \pm Z_{\alpha/2} \sqrt{\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}} = 2.83 \pm 1.96 \sqrt{\frac{19.1}{552} + \frac{16.4}{744}} = [2.36, 3.29]$$

假设检验

□ 假设检验——案例4

There is interest in testing whether the repeal of the NMSL had an effect on the speed variances rather than mean speeds. The test is conducted at the 10% significance level. Descriptive statistics show that s_b^2 and s_a^2 , the standard deviations for the before and after the repeal periods, are 16.4 and 19.1, respectively. The parameter of interest is variance and the competing hypotheses are

$$H_0: \sigma_a^2 - \sigma_b^2 = 0$$

$$H_a: \sigma_a^2 - \sigma_b^2 \neq 0$$

$$F_{(n_1-1, n_2-1)}^* = F_{(522, 744)}^* = \frac{s_1^2}{s_2^2} = \frac{19.1}{16.4} = 1.16$$

显著性0.01条件下的检验统计量关键值是1，拒绝无效假设；即如果前后方差相等，观测到这个差别的可能性较低。



假设检验

□ 参数 vs. 非参数

- 参数检验有诸多假设：分布正态性，数据连续性，样本量大等
- 非参数检验假设较少，使用数据信息量也较少
- 同时运用两种检验时，非参数检验犯II类错误的可能性较大
- 参数检验：近似问题的精确解
- 非参数检验：精确问题的近似解

□ 何时使用非参数检验？

- 参数检验假设不满足
- 研究对象不是总体参数



假设检验

□ 卡方检测 (chi-square test)

- 提出无效假设和备择假设
- 在无效假设的条件下，基于某统计模型（理论分布、经验分布等）计算事件期望发生频率（所有类别均大于等于5）
- 把观测数据分门别类
- 计算卡方检测统计量
- 与卡方分布关键值比较

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$



假设检验

□ 卡方检测：列联表 (contingency table)

TABLE

General Layout of a Contingency Table

Second Classification Category	First Classification Category			Total
	1	.	j	
1	C_{11}	.	.	R_1
i	.	.	C_{ij}	R_i
Total	C_1	.	C_j	n

H_0 : The two classification variables are statistically independent.

H_a : The two classification variables are not statistically independent.

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(C_{ij} - E_{ij})^2}{E_{ij}} \quad E_{ij} = \frac{R_i C_j}{n} \quad df = (r - 1)(c - 1)$$

